

The holding vs ordering cost trade-off (preview of lot sizing)

Part 6 · Show me the money — solution report with data, method, results and charts

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Item **Inverter PCB** — $D = 4,800/\text{yr}$, $v = 410$, $A = 2,630$ per order, $r = 37\% \rightarrow h = 151.70$ per unit-year.

$$\text{TRC}(Q) = A \cdot D/Q + h \cdot Q/2$$

408 units

Cost-minimising Q (numerical)

11.8

Orders per year

61,888/yr

Minimum total relevant cost

30,941 vs 30,947

Ordering = holding at optimum?

262–635

Cost within +10 % for Q in

Answering 'how large should the order be?'

Minimum total cost where ordering and holding costs balance — and the curve is flat near the optimum

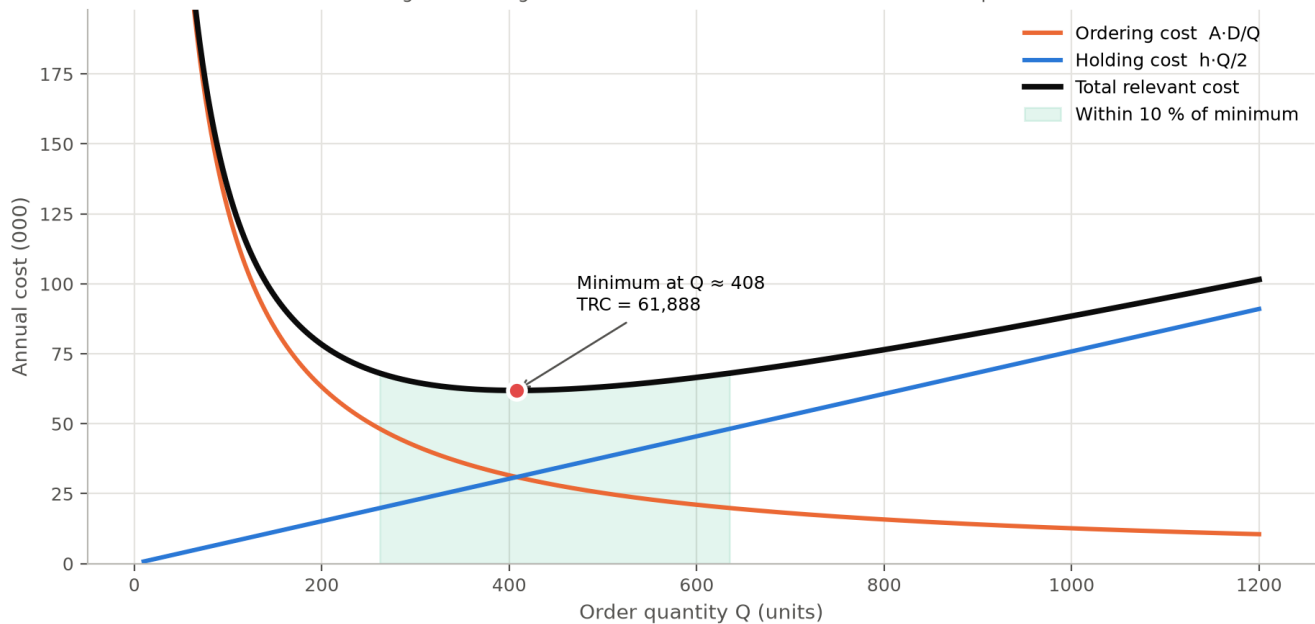


Figure 1.1 — Annual ordering, holding and total relevant cost vs order quantity.

Cost of alternative order quantities.

Q	orders per year	avg cycle stock	ordering cost	holding cost	total cost	pct above min
50	96.0	25	252,480	3,792	256,272	+314.1%
100	48.0	50	126,240	7,585	133,825	+116.2%
200	24.0	100	63,120	15,170	78,290	+26.5%
408	11.8	204	30,941	30,947	61,888	+0.0%
400	12.0	200	31,560	30,340	61,900	+0.0%
800	6.0	400	15,780	60,680	76,460	+23.5%